



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO M.A.N FUELS AND CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

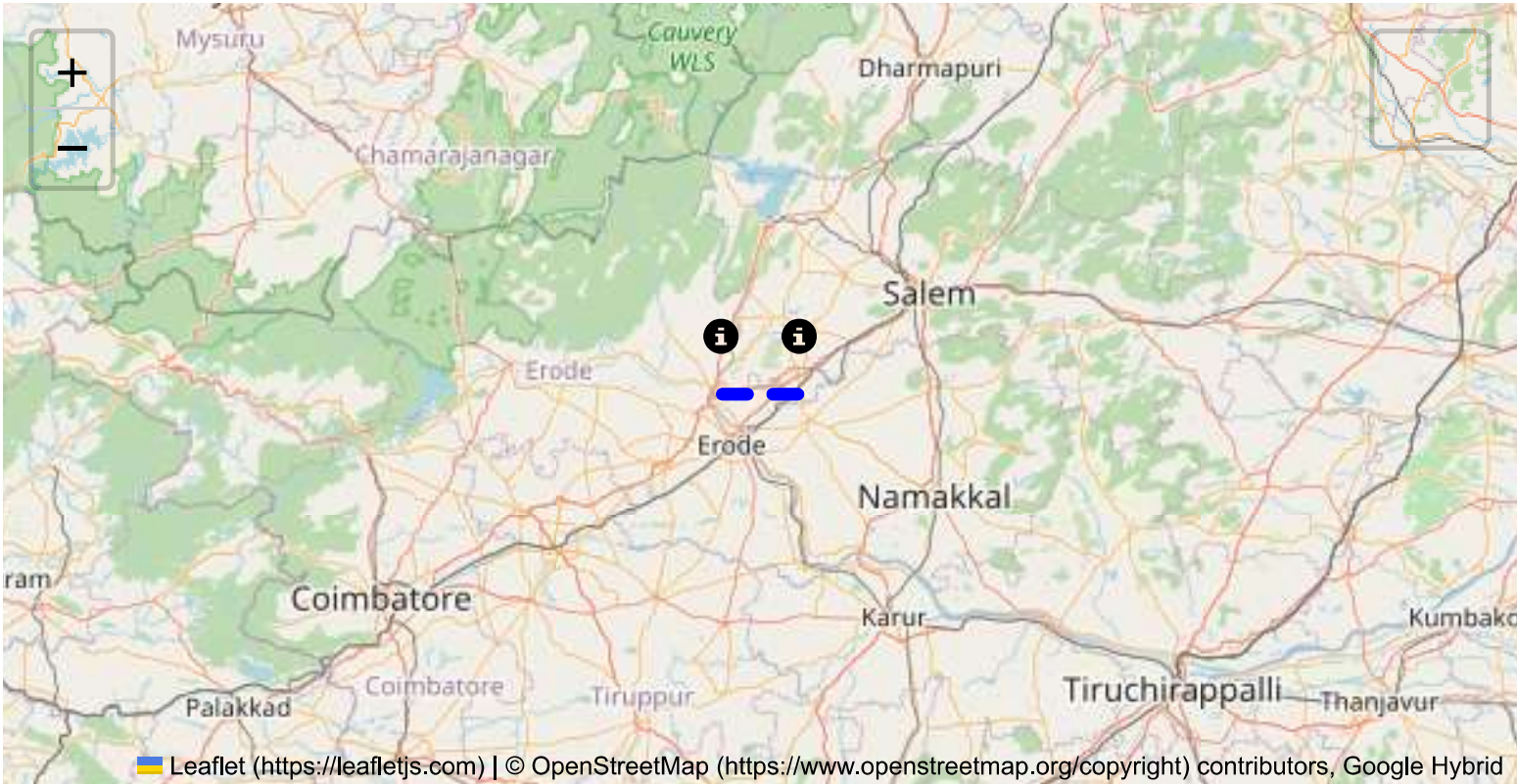
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

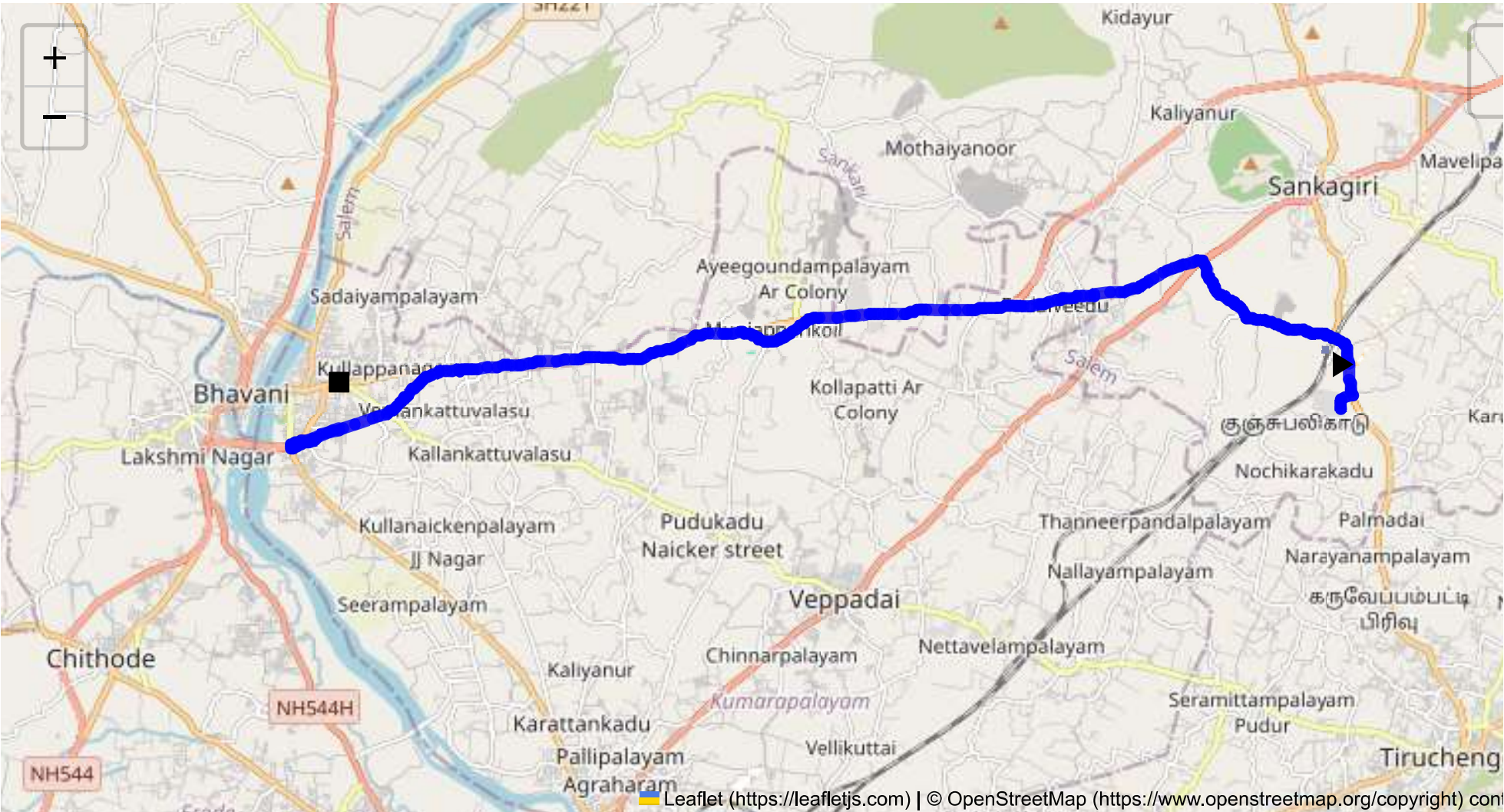
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 23.90 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.6 hours
Start: (11.4381, 77.8734)
End: (11.434815, 77.699218)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri (CVQF+23W) to Komarapalayam (CMMX+VJ9) spans approximately 23.90 kilometers and typically takes around 30 to 40 minutes for heavy vehicles carrying hazardous materials. It primarily follows state highways known for being arterial roads with moderate connectivity.

Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate with a mix of dry and wet seasons. The route can encounter:

- **Monsoon Season (June to September):** Heavy rains and thunderstorms may lead to flooding or slippery roads.
- **Summer (March to May):** High temperatures often exceeding 35°C, which might affect vehicle performance.
- Ensure regular weather checks, especially during the monsoon, to anticipate natural hindrances.

Traffic Patterns

- **Peak Hours:** Typically observed during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) commutes, especially in the vicinity of towns like Sankari and Komarapalayam.
- **Congestion-Prone Areas:** Approaches to town areas and intersections with major state highways often see traffic buildups.

Road Quality and Infrastructure

- The roads along this route are generally well-maintained with occasional potholes that need caution.
- Some sections may have narrow lanes, requiring careful navigation for heavy vehicles.
- Watch for unexpected road work or repair zones, which may not always be adequately signposted.

Alternative Routes for Emergencies

- **Route A via SH 188:** Offers a bypass around major congestion points but may have less signal coverage.
- **Local Roads:** Many small roads connect with the main route, allowing detours, but ensure they are suitable for heavy vehicles.

Local Regulations for Hazardous Material Transport

- Strict adherence to state regulations for the transport of hazardous materials, including proper labeling and documentation.
- Night-time travel restrictions may apply for highly volatile goods; check local regulations before planning the journey.

Historical Incidents

- Few recorded incidents involving heavy vehicles along this specific route.
- Historical weather-induced road damage sometimes leads to traffic delays during monsoons, emphasizing the need for vigilance and pre-trip planning.

Environmental Considerations

- Eco-sensitive zones have less prevalence, but care must be taken near agricultural zones to minimize disturbance.
- Waste management protocols should be strictly followed to prevent local contamination.

Communication Coverage

- Generally stable in town areas but potential dead zones exist in certain rural segments.
- Equip vehicles with proper GPS and consider satellite phones or CB radios for ensured communication.

Estimated Emergency Response Times

- **Urban Areas (like Sankari):** Roughly 20-30 minutes for emergency services.
- **Rural Segments:** Could extend to 30-45 minutes depending on proximity to major towns.

Overall Summary of Risk Assessment

- **Overall Risk Level:** Moderate
- **Primary Concerns:** Weather-induced hazards during monsoons, potential traffic congestion, and road infrastructure challenges.
- **Recommendations:** Pre-route weather assessments, adherence to transport regulations, prepared communication tools, and knowledge of alternative routes.

This assessment provides a general overview, and the truck drivers should be briefed regularly with updates to ensure safety and compliance with the latest local advisories.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.46307, 77.84941	30 KM/Hr
5	Turn	Medium	11.46318, 77.84913	30 KM/Hr
6	Turn	High	11.45517, 77.81345	15 KM/Hr
7	Turn	High	11.45521, 77.81343	15 KM/Hr
8	Turn	Medium	11.44467, 77.72092	30 KM/Hr
9	Turn	Medium	11.43994, 77.71111	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.43982, 77.71084	30 KM/Hr
11	Blind Spot	Blind Spot	11.43297, 77.69543	10 KM/Hr
12	Blind Spot	Blind Spot	11.43283, 77.69549	10 KM/Hr
13	Turn	High	11.43134, 77.69123	15 KM/Hr
14	Turn	High	11.43188, 77.69112	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital	11.4525596, 77.7749426	30 km/h	Medium
4	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
5	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
6	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
7	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium

Route Photos of Risky Spots